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Final External Package-Residual Re-audit – Paper II v1.2, corrected in place

Specification: FINAL_RESIDUAL_AUDIT_REQUEST_SPEC.md **Specification SHA-256:**

2eadd655d6dd9f1d96127c979ba43a0b2d251da7478cd84ba4bb1868e38cb683

Request ID: PAPER_II_EXTERNAL_PACKAGE_RESIDUAL_v1.0 **Run:** run_2026-08-21_v1.2_pkgfix **Audit class:** independent external residual adversarial re-audit **Audit date:** 2026-08-21

1. Executive verdict

PASS

EXT-PII-M-001 is independently **closed**. The 245-file target inventory verifies, **including the canonical manifest hash**, all nine protected anchors match, the package regression found **no new blocker, major or minor**, reused evidence satisfies byte identity, and EXT-P2-I-001 is preserved as a NOTE.

Section 7 condition	Met
1. EXT-PII-M-001 independently closed	yes
2. 245-file inventory and every manifest entry verify	yes
3. all protected anchors match	yes, 9/9
4. package regression finds no new blocker, major or minor	yes, 0/0/0
5. reused evidence satisfies byte identity	yes
6. EXT-P2-I-001 preserved as a NOTE	yes

This audit is not human peer review and does not prove global novelty.

The previous PASS_WITH_RESIDUALS run is not overwritten or relabelled.

2. Target freeze – the manifest reproduced exactly

Quantity	Recomputed	Declared	Result
files	245	245	MATCH
bytes	3,940,779	3,940,779	MATCH
manifest SHA-256	see below	same	MATCH

The recomputed manifest hash, identical to the declared value:

4b41f7e2e9415ca55514c6997dc6bff4e952b830282f46012c5214b04f688c1e

Worth stating plainly, because it contrasts with the Paper I package-fix run: this specification **publishes the manifest algorithm** – UTF-8, LF-only lines of the form `<file_sha256><two spaces><target-relative POSIX path>\n`, sorted case-insensitively by path, then SHA-256 over the exact manifest bytes. Because the algorithm is published, an external party can reproduce the summary value, and this auditor did, on the first attempt. In the Paper I run the corresponding aggregate was unreproducible under sixteen scope-by-convention combinations and had to be recorded as an unverifiable control. That is the difference a published algorithm makes.

Every file hash, byte count and path was recomputed before analysis. **No file-entry mismatch.** No manifest-summary mismatch either, so the two are reported together here rather than separately.

3. Protected anchors – 9 / 9

Each of the nine declared anchors was recomputed from the frozen target and **matched**. Recomputed values, in declaration order:

7215e14bbea8ab2bf208dcdd1efa050cd2b72c997eee2efe504a1e6817c68882	English Markdown
d0d1df05eb267a51db2ccc100dd9725dcde9b03dbb95c8a730742e357eb0f4dc	Spanish Markdown
bb5f76c3ce56dbb0bff11242a3a8787f9c8ba3d9f0ad23973fc2f26cc5fc3cf0	English LaTeX
d3f0c6301a48d6553ebad222fa685f152119cb61b5efd3e8be55e389f9d606ae	Spanish LaTeX
d05c4cab1262357fddd21e4aab399bdb92d5bcf139172897c80595e781049052	English PDF
d525d02a6e911cb23f7e1f28e1de7648441eccea6de206e76e5321161c86c2db	Spanish PDF
ee2d05cc40d943ca92f8f7bf3e5dd83c2692518ddea5e2ca4f7686ccb1ac3895	Lean v1.2 archive
1e7afd3e9394bf83beb7e33ce19ff5227072fcd6b0eb3fd21e571329564e3ded	Previous external report
779c8453e0ec3666bd8a7564d2e1251b6f2acf586815cd43470c202fa657eb13	Internal residual-audit ZIP

The previous external report was located at `run_2026-08-21_v1.2/30_REPORT/FINAL_AUDIT_REPORT.md`.

All six publication artifacts and the Lean archive are **unchanged**, which is the precondition for the evidence reuse in Section 7.

4. Control A – EXT-PII-M-001 is closed

4.1 The six corrected integrity documents

All six declared anchors match, and every one is **LF-only**:

All six live in `04_integrity/`.

File	Anchor	Result
INITIAL_SOURCE_SHA256.txt	1d5f8d91...0323	MATCH, LF
INITIALIZATION_DIFF.md	4b88e306...0c2c	MATCH, LF
README.md	d7ed1d33...0e85	MATCH, LF
CURRENT_TARGET_SHA256.txt	79fd6542...a315d	MATCH, LF
SEMANTIC_INTEGRITY_REPORT_v1.2.md	5c9d6dc4...d817e	MATCH, LF
EXTERNAL_AUDIT_V1.2_	a004b4fe...0857	MATCH, LF
RESIDUAL_MATRIX.md		

4.2 Each bullet of the control, checked independently

Requirement	Result
INITIAL_SOURCE_SHA256.txt is LF-only and all three entries resolve from the package root and match	PASS – LF, exactly 3 entries, 3/3 resolve and match, 0 missing
it contains no absent PAPER_II_preprint_draft_v1.1.md path	PASS – no preprint_draft_v1.1 path present
CURRENT_TARGET_SHA256.txt is LF-only and all nine entries resolve and match	PASS – LF, exactly 9 entries, 9/9 resolve and match
04_integrity/README.md describes v1.2, not a pending v1.1 workspace	PASS – it opens “Integrity records – Paper II v1.2”, the string “Pending for v1.1” is gone , and there is no reference to a superseded/ directory the package does not contain. It states explicitly that it “replaces the stale v1.1 integrity records identified by external finding EXT-PII-M-001”.
INITIALIZATION_DIFF.md documents v1.1 to v1.2 and claims no protected mathematical change	PASS – it names the v1.1 English Markdown hash 361802b5...ded6 and the v1.2 hash 7215e14b...8882, records that the historical package is retained outside the active target, and asserts no protected change
the semantic-integrity report and residual matrix agree with the actual hashes and external findings	PASS – both match their declared anchors and reference EXT-PII-M-001 correctly
no sidecar anywhere in the target fails by content	PASS – see 4.3

The v1.2 defect was precisely that this directory was stale v1.1 content whose INITIAL_SOURCE_SHA256.txt named a manuscript absent from the package, so one of its two entries could not resolve. **All three entries now resolve and match, and the directory describes the target it ships with.**

4.3 Every sidecar in the target: 40 of 40 verify

40 sidecars were verified by content – 18 LF, 22 CRLF, none mixed. **All 40 verify with zero mismatches and zero missing files.**

One initially appeared to fail: the copy of PAPER_II_preprint_draft_v1.2_SHA256.txt under the internal run’s 20_EVIDENCE/R5_ARTIFACTS/results/. It is a **byte-identical copy** of the real manuscript sidecar (fa0d1501...e962 for both), placed in the internal run’s evidence directory, whose entries are bare filenames and therefore resolve only from 01_manuscript/, its documented base. Resolved there it verifies **6/6**. The initial flag was this auditor’s base-directory resolver not including that directory. Recorded so a tool artifact is not left looking like a target defect.

Control A verdict: PASS. EXT-PII-M-001 is closed.

5. Control B – the internal residual PASS was inspected, not trusted

The internal report hashes both match: Markdown 3fac92e6...0fe0, PDF dfdfef4f6...6761.

Gate	Spec expectation	What the raw evidence shows
R0	17/17 controls, canonical manifest algorithm	status: PASS, checks_passed: 17, checks_total: 17, and a canonical_manifest_algorithm field. Their inventory scope is 200 files / 3,417,038 bytes – narrower than this run’s 245, because it excludes their own residual run; consistent, not contradictory.
R1	58/58 common static controls	status: PASS, checks_passed: 58, and audit_class: INTERNAL_NOT_EXTERNAL, lean_rerun: False – an honest self-label

Gate	Spec expectation	What the raw evidence shows
R2	exit-zero exact regression over the unchanged Markdown hash	status: PASS, keyed to target_sha256: 7215e14b...8882, the unchanged EN Markdown. Real instance counts, not a bare label: 195 complete-split LP instances, 5,000 exact integer maximizations, 6,667 argmax level checks, 139 chordal atlas graphs for $n \leq 6$, 931 vertex-copy non-edge pairs, 22 terminal-property graphs. Their own scope note calls bounded computation “corroborating evidence”, which is the correct posture.
R3	zero exact or near duplicate blocks in all six artifacts	status: PASS, threshold 0.985 on normalized paragraphs of ≥ 180 characters, zero exact and zero near duplicates in every artifact
R4	copied external logs byte-consistent with the previous external run; build exit 0, 8,063 jobs, 16 axiom surfaces	PASS, and verified at the byte level against this auditor’s own artifacts – see 5.1
R5	all six publication artifacts match the prior anchors	status: PASS_UNCHANGED_ARTIFACTS, manuscript_artifacts_changed: False, pdfs_rebuilt: False, figures_changed: False
R6	report compilation evidence, gate manifests, ZIP members and sidecar	ZIP 779c8453...eb13 matches its own sidecar and the specification anchor; 42 members, all byte-identical to the loose files

5.1 R4 is the one gate this auditor could check against its own evidence, and it holds

Specification Section 5 says R4 consists of external raw logs **copied** into the internal run. Those logs are this auditor’s own output from run_2026-08-21_v1.2, so byte-consistency is checkable rather than a matter of trust. Comparing by SHA-256:

Copied file	Byte-identical to the external original
01_lake_update_cache_get.log	yes
02_lake_build_protocol_9_2.log	yes
03_FreezeAxioms_run.log	yes
EXTERNAL_H_AUDIT_RECORD.md	no – their own new record, correctly not a copy
summary.json	no – their own new summary

Read from **their** copies, not from the originals:

- Build completed successfully (8063 jobs)., EXIT_BUILD=0, 0 error lines;
- 16 axiom surfaces, EXIT_AXIOMS=0, 0 sorryAx and 0 project axioms;
- 2 surfaces carry the strictly smaller footprint [propept, Quot.sound]: PaperII.phiTau_max_le_perI_bound and SimpleGraph.IsChordal.comap – exactly the two this auditor identified independently in the external run.

Their summary.json records lean_executed: False alongside external_build_exit: 0 and external_build_jobs: 8063. That is an accurate and honest labelling of reuse rather than execution.

Control B verdict: PASS. The label is backed by raw evidence, and the one gate that could be cross-checked against independent artifacts matches byte for byte.

6. Control C – general package regression

6.1 Hygiene: clean

Check	Result
any tmp directory	none
compiler scratch (.aux, .toc, .out, .fls, .fdb_latexmk, .synctex, .nav, .snm, .bbl, .blg, .idx, .ilg, .ind, .lof, .lot, .spl, .bcf, .run.xml)	none
zero-byte files	none
stray \$o or any \$ in a filename	none
unexpected hidden files	none

6.2 Authorized delta: exact, with zero deletions

Delta	Count	Classification
changed	6	exactly the six authorized files (the three 04_integrity/ documents named above, plus DRAFT_METADATA.yml, DRAFT_NOTES.md and the root README.md)
added	48	the 3 authorized integrity files, plus 45 files of the authorized internal residual run (under 01_INTERNAL_AUDITS/, directory residual_run_2026-08-21_v1.2_pkgfix)
removed	0	no deletion, as required
unauthorized	0	–

197 files -> 245 files. No manuscript or formal change, so evidence reuse is not blocked.

6.3 Self-containment, stale references, status

- **Manuscript self-containment: 0 violations** in both languages. Neither manuscript references v1.0.1, a correction matrix, a residual matrix, an audit report, or any EXT-PII-* / EXT-P2-* finding identifier.
- **Stale v1.0.1 references: 3, all investigated and none a reappearance.** All three are in 02_validation/00_intake/PAPER_II_preprint_v1.1.8_near_final_editorial_en.md, a historical intake document from the v1.1.8 era that legitimately contains the old references because it *is* the old document. It appears in neither the changed nor the added list, so it is byte-unchanged from the previous external freeze – pre-existing, not reintroduced. The specification asks that no stale reference **reappeared**; none did.
- **Status consistency: exact.** Both README.md and DRAFT_METADATA.yml carry the literal status INTERNAL_PACKAGE_RESIDUAL_PASS_EXTERNAL_RESIDUAL_PENDING, and DRAFT_NOTES.md states “a final external residual re-audit is pending”. DRAFT_METADATA.yml also distinguishes local_build: PASS_RECORDED_8061_AND_8032_JOBS_NOT_RERUN from independent_reproduction: PASS_EXTERNAL_CLEAN_ROOM_8063_JOBS – recorded evidence and independent reproduction kept apart, correctly.
- **No truncation, placeholder or malformed final report** was found.

Control C verdict: PASS.

7. Reused evidence – REUSED_BYTE_IDENTICAL_EXTERNAL_EVIDENCE

All nine protected anchors match, so the prior external mathematical, formal, citation, novelty, bilingual, duplicate and rendered-PDF evidence is reused. **These gates were not rerun.** Specifically not rerun: the 30-minute Lean build, the full mathematical falsification, and the 47-page rendered analysis.

Reused results, from run_2026-08-21_v1.2:

Reused item	Recorded result
clean Lean build	exit 0, 30m05.109s, Build completed successfully (8063 jobs), zero errors

Reused item	Recorded result
axiom footprint	16 declarations; 14 with <code>[propept, Classical.choice, Quot.sound]</code> , 2 with the smaller <code>[propept, Quot.sound]</code> , which the manuscript’s Table 5 records accurately; no <code>sorryAx</code> , no project axiom
escape hatches in active code	0 across 42 project files
headline falsification	19,048 chordal graphs enumerated exhaustively for $n \leq 6$; <code>max Phi_tau = floor((2n+1)^2/24)</code> exact at every n ; always attained by a complete-split graph
vertex-copy inequality	251,085 instances, 0 violations, minimum defect 0
the identity $\text{nu}3\text{star} = \text{tau}3\text{star}$	270,133 graphs, two separate linear programs, 0 mismatches
arithmetic corollaries	all four confirmed over n in $[-20000, 20000]$, including negative n ; the $n \geq 1$ hypothesis on the Paper I comparison shown necessary
bilingual and rendered PDF	protected content identical; 47 pages, 0 anomalies; the Paper I duplication defect class absent
citations	[1] verified verbatim against the publisher record; [3] Blair-Peyton retrieved

Prior scope limitations are preserved, not quietly dropped:

- termination of the repeated-copy process was **not** independently verified;
- the discrete-convexity lift from single vertices to clone classes was **not** verified;
- the two-variable orbit reduction on $S_{\{p,q\}}$ was tested only through its consequences;
- exhaustive enumeration reached only $n = 6$;
- Lean modules outside the seven protocol targets were never compiled;
- the PDFs were not independently recompiled from the delivered TeX;
- **the prior Paper II audit used a single reasoning context.** No adversarial challenger was run for Paper II, in that run or this one. On Paper I the challenger found four Spanish duplications the primary auditor had missed, so this is a real limitation and not a formality.

8. Findings

ID	Severity	Status	Summary
EXT-PII-M-001	MINOR	CLOSED	stale v1.1 integrity directory; now describes v1.2, all three sidecar entries resolve, six corrected documents match their anchors
EXT-P2-I-001	NOTE	OPEN	the official Erdős Problems page returns HTTP 403 to this auditor. Not repaired , and not represented as repaired: no direct authoritative access was obtained. The substantive open-status claim is supported by the verified primary EOZ source, whose abstract states “It is unknown whether this many cliques will always suffice.”

No new blocker, major or minor.

9. What this audit does not establish

- It does **not** establish that the headline theorem is true. The prior external run’s exhaustive falsification over 19,048 chordal graphs and 251,085 copy instances, plus an independent formal reproduction, failed to falsify it. That is the strongest claim the evidence supports.
- It is **not** human peer review.
- It does **not** prove global novelty. The prior Gate K conclusion was an evidence-bounded negative search result and remains exactly that; no institutional bibliographic database, citation-graph traversal or non-English search was available in any run.
- It does **not** verify termination of the copy process or the clone-class lift.
- It does **not** re-derive the Lean development in this run; that evidence is reused under byte identity.
- Paper II has never been audited by an adversarial challenger.

10. Signature

Item	Value
Auditor	Claude Opus 5 (<code>claude-opus-5</code>), Anthropic
Configuration	primary auditor; no challenger, for this run or the prior Paper II run
Audit date	2026-08-21
Specification	<code>FINAL_RESIDUAL_AUDIT_REQUEST_SPEC.md</code> , SHA-256 <code>2eadd655...b683</code>
Target	<code>preprint_draft_v1.2</code> corrected in place, 245 files, 3,940,779 bytes, manifest <code>4b41f7e2...8c1e</code>
Verdict	PASS – EXT-PII-M-001 closed; 0 new blocker/major/minor; EXT-P2-I-001 preserved as a NOTE

The previous `PASS_WITH_RESIDUALS` run is not overwritten. It stands against its own hashes as the record of the state that preceded this correction.